

Assignment – 4.4

Q - 1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

RAM Reseating:

1. Shut down the computer and disconnect the power.
2. Open the case and locate the RAM slot.
3. Push the locking clips outward and remove the RAM.
4. Check the RAM and slot for dust or damage.
5. Align the notch and reinsert the RAM firmly.
6. Ensure the clips lock the RAM in place.
7. Close the case and reconnect the power.
8. Start the computer and check if the RAM is detected.

Issues observed: No physical damage; RAM was properly reseated and detected successfully.

Q - 2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix. Hint: Think about display issues, driver checks, and physical connections.

GPU Troubleshooting Checklist:

1. **Check the problem** – No display, flickering, or game crashes.
2. **Check connections** – Make sure the GPU is properly connected.
3. **Check display cable** – Ensure HDMI/DisplayPort is connected correctly.
4. **Check drivers** – Update or reinstall the GPU driver.
5. **Check temperature** – Make sure the GPU is not overheating.
6. **Test the GPU** – Run a game and check its performance.
7. **Confirm the fix** – Restart the PC and check if everything works properly.

Q - 3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Storage Issue:

1. Turn off the computer.
2. Unplug the SSD/HDD.
3. Turn on the computer.
4. **Before reconnecting:** BIOS may show “**Drive Not Detected**” or “**No Boot Device Found.**”
5. Turn off the computer and reconnect the drive.
6. Turn on the computer again.
7. **After reconnecting:** BIOS detects the drive and the computer boots normally.
8. **Result:** The storage problem was fixed by reconnecting the drive.

Q - 4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Random PC Restart – Troubleshooting:

➤ **Component to check first: RAM**

Reason: Faulty or poorly connected RAM can cause system crashes and random restarts.

➤ **Diagnostic Steps:**

1. Turn off the PC and unplug it.
2. Open the computer case.
3. Remove the RAM.
4. Clean the RAM contacts gently.
5. Reinsert the RAM properly.
6. Turn on the PC and check.
7. Run **Windows Memory Diagnostic**.
8. If there are two RAM sticks, test them one by one.

Conclusion: If the restarts stop, the RAM was likely the problem.